Image Interpolation Using Non-adaptive Scaling Algorithms for Multimedia Applications—A Survey

C. H. Neetha, C. John Moses, and D. Selvathi

Abstract Image interpolation is a technology of maintaining the quality of digital images by introducing new high-quality pixels. The interpolation schemes are used in several image processing applications like computer graphics, reconstructing medical images, producing high resolution in still images, and in television pictures as well as processing remote sensing images. This proposed survey deals with the study of different non-adaptive image interpolation techniques used for multimedia applications. This study analyzes the characteristics of different non-adaptive image interpolation algorithms in terms of peak signal to noise ratio (PSNR) and structural similarity index measure (SSIM). From the review on interpolation schemes, this work identifies that several low-complexity conventional non-adaptive image interpolation schemes outperform the recently developed non-adaptive image interpolation algorithms in terms of PSNR. Further, it suggests that for any decisive applications, the conventional non-adaptive interpolation schemes can also be used to maintain the quality of picture.

Keywords Up-scaling · Sampling · Convolution · Resolution · Bilinear · Bi-cubic

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V. Komanapalli et al. (eds.), *Advances in Automation, Signal Processing, Instrumentation, and Control*, Lecture Notes in Electrical Engineering 700,
https://doi.org/10.1007/978-981-15-8221-9_141

1 Introduction

In recent years, different methods of non-adaptive image interpolation algorithms have been developed for multimedia applications. Image interpolation provides high-quality image by zooming/sampling or scaling/reconstruction [1]. Since the manufacturing cost of digital cameras increases, while providing optical zooming and while increasing the capability of capturing megapixel, to keep the manufacturing cost at a reasonable level, the manufacturers produce digital cameras with the capacity of performing digital zooming. The digital image zooming technology performs interpolation spatially to increase the spatial resolution by completing the missing pixel from the neighboring pixels, and it also preserves spectral information of the digital image [2].

Image interpolation is used to magnify still images or video sequences by zooming images on digital cameras or by producing high resolution video materials on high definition television (HDTV) [3]. Image interpolation is one of the methods of achieving super resolution (SR) image from low resolution (LR) image. The various applications of SR images are found in criminal investigation, video surveillance, and digital entertainment [4]. Interpolation-based image SR is computationally efficient [5]. Interpolation is used for converting standard definition television (SDTV) to HDTV [6]. A TV scalar is used to produce high resolution videos without flickering artifacts from the low-resolution input by using double interpolation [7]. Image interpolation is used to restore the missing pixel areas in digital images by removing unnecessary objects, by restoring corrupted old films or by error enhancement of video communication [8]. A tetrahedral interpolation process is used in gamut mapping for improving color reproduction quality of digital television [9]. A polynomial-based image interpolation is used to enlarge the image and to reduce the block effect of image/video coder [10].

The remaining part of this paper is organized as follows. Section 2 describes the different interpolation techniques used on multimedia applications. Section 3 gives the comparative analysis, and Sect. 4 provides the conclusion and future scope.

2 Non-adaptive Image Interpolation Methods

The non-adaptive interpolation is preferred in many multimedia devices due its less computational effort when compared with adaptive image interpolation. Non-adaptive interpolation treats all pixels equally. This section provides the details of various non-adaptive image interpolation algorithms in three different categories such as linear/non-linear, spline, and convolution-based interpolation.

A linear interpolation for real-time digital image scaling is proposed by [11]. This method achieves highest PSNR as 33.33 dB. But the similar algorithm presented by Kim et al. [12], achieves only 29.74 dB. Further, Lin et al. [13] proposed an extended linear interpolation by using 16 weighting coefficients provided by 16 surrounding pixels of the input image. This algorithm concentrates on reducing computational complexity and provides the same PSNR as the previous algorithm [11]. Chen [14] proposed a non-adaptive bilinear interpolation with pre-filters like clamp and sharpening filters. These filters are utilized to reduce the image artifacts like blurring and aliasing. This work provides a highest PSNR of 28.80 dB by utilizing clamp filter. Adder-based stepwise linear interpolation (ABSI) is proposed by [15]. This method provides a low-complexity interpolation by using simple addition operation. This method achieves a highest PSNR as 32.15 dB. Recently, Acharya and Sukader [16] presented a non-linear fuzzy-logic-based image up-scaling. This system uses a fuzzy composite scheme based on inverse-modeling approach to restore the high frequency details of the up-scaled images. This method undergoes pre-processing to boost-up very high frequency components using recursive Laplacian operator. This method provides highest PSNR as 35.56 dB and universal quality index (UQI) as 0.9922. Further, Moses et al. [17] proposed a high-performance ABSI by using combined filter as pre-filter. This method achieves a highest PSNR as 24.12 dB and SSIM as 0.9352 for interpolating Kodak color images. By comparing the experimental results of this method with [15], this method provides higher PSNR for Kodak image interpolation.

Cubic-spline interpolation (CSI) with cross zonal filter (CZF) is proposed by different researchers such as Wang et al. [18] and Hong et al. [19]. Wang et al. [18] decimates the input image with the ratio 4:1 and 16:1, respectively, and interpolates the image by using the ratio 1:4 and 1:16, respectively. This algorithm provides a highest PSNR as 35.08 dB. Hong et al. [19] use six points cubic convolution interpolation (CCI) and with CZF. This system compares the quality of the interpolated image by using different values of optimal parameter (α). This CSI provides 37.9 dB by using $\alpha = -1$. This CSI achieves greater PSNR when compared with other type of CSI [18]. Hong et al. [20] proposed another new CSI to resample the discrete image data by using least-square method. This method achieves highest PSNR as 34.73 dB. But the PSNR value of this method is lesser than the previous CSI methods [18, 19]. The traditional polynomial spline (PS) is combined with low-pass filter (Sinc filter) to improve the quality of the interpolation kernel by Madani et al. [21]. The highest PSNR value on the up-scaled image is 38.02 dB. This PSNR value is higher than the other spline interpolation like CSI with CZF [18, 19] and the recent CSI [19].

A modified convolution-based image interpolation is proposed by Sinha et al. [22]. This algorithm achieves a highest PSNR as 29.87 dB, SSIM as 0.790. Qiao et al. [23] proposes a single image interpolation for digital photography. This method exploits a variation framework with time varying parameters. This method provides highest PSNR as 37.86 dB and SSIM as 0.9590. Polynomial convolution interpolation is proposed by Tsai and Teng [24] by using second-order polynomial approximation. This algorithm uses nine neighboring pixels to interpolate the given image. This interpolation algorithm achieves a highest PSNR as 38.80 dB. A modified cubic convolution interpolation is proposed by Chae et al. [25] to optimize the local property of image data and to reduce the information loss during the scaling process. This algorithm uses bi-cubic interpolation technique with optimized turning parameter ($\alpha = -1$ to 0.5) of the kernel and interpolation filter. This method achieves a highest

PSNR as 39.15 dB. Huang et al. [26] presented another convolution-based image interpolation with error correcting scheme. This avoids checkerboard effect and blurring caused by conventional algorithms. The highest PSNR value of this algorithm is 47.50 dB. A first-order polynomial convolution interpolation (FOPCI) is proposed by Lin et al. [27]. The kernel of this interpolation technique uses first-order polynomials, and it also approximates the ideal sinc-function. The highest PSNR of this method is 49.53 dB. Block region of interest-based image up-scaling method using Sinc interpolation is proposed by Li and Yu [28]. This algorithm uses partitioning scheme to subdivide the target image into blocks on the basis of region in the source image. Recently, Karaca and Tunga [29] proposed an interpolation-based image inpainting to fill missing regions of input image by using intensities of pixels in the surrounding area. This method generates inpainting using high dimensional model representation (HDMR) with Lagrange interpolation. This method gives highest PSNR as 44.33 dB. Moses et al. [30] presented T-model clamp filter-based Lagrange image interpolation which provides a highest PSNR as 28.99 dB and SSIM as 0.9801. Further, bi-cubic interpolation examined by Vargas et al. [31] outperforms others with 39.69 dB and 0.9781 SSIM on interpolating magnetic resonance imaging (MRI) images.

3 Comparative Analysis

This section provides performance comparison of different non-adaptive image interpolation algorithms based on their quantitative measures such as PSNR and SSIM. Figures 1, 2, and 3 show the comparison of PSNR of different interpolation category such as linear/non-linear, CSI, and convolution-based. Further, Table 1 shows the overall comparison of PSNR and SSIM of different non-adaptive image interpolation algorithms.

Figure 1 demonstrates the comparison of different interpolation linear/non-linear interpolation algorithms based on the PSNR such as linear interpolation (LI) [11], extended linear interpolation (ELI) [13], bilinear with sharpening filter (SLSF) [14], ABSI [15], non-linear interpolation (NIL), [16] and ABSI with combined filter (ABSICMF) [17].

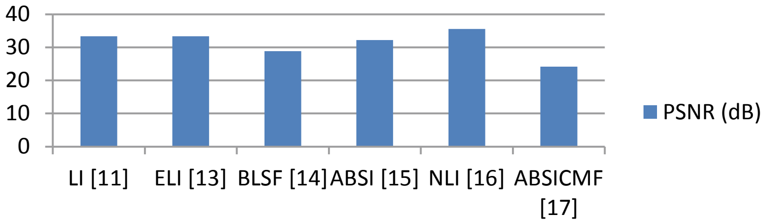


Fig. 1 Comparison of different linear with non-linear interpolation algorithm based on PSNR

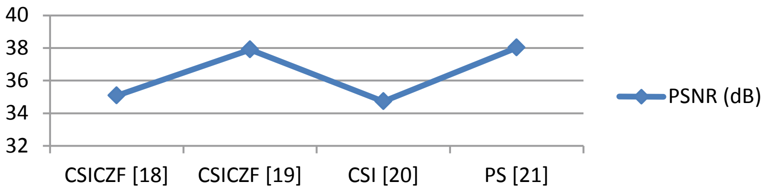


Fig. 2 PSNR comparison of different spline interpolation methods

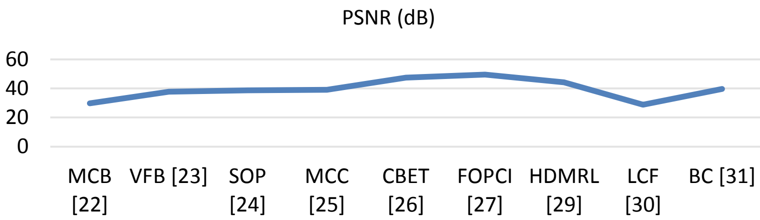


Fig. 3 Comparison of PSNR various convolution-based interpolation methods

Table 1 PSNR and SSIM comparison of different non-adaptive image interpolation algorithms

Interpolation method	PSNR (dB)	SSIM
Combined filter-based ABSI [17]	24.32	0.935
Bilinear with pre-filters [14]	28.80	N/A
Clamp filter-based Lagrange [30]	28.99	0.980
Modified interpolation [22]	29.87	0.790
Linear interpolation [11]	33.33	N/A
Cubic-spline [20]	34.73	N/A
Cubic-spline with cross zonal filter [18]	35.08	N/A
Extended linear [13]	35.29	N/A
Non-linear fuzzy logic-based [16]	35.56	0.992
Variation framework-based [23]	37.86	0.959
Cubic spline with cross zonal filter [19]	38.00	N/A
Polynomial spline [21]	38.02	N/A
Second-order polynomial [24]	38.80	N/A
Bi-cubic with pre-filter [25]	39.15	N/A
Bi-cubic [31]	39.69	0.9781
Lagrange with HDMR [29]	44.33	N/A
Convolution-based with error theorem [26]	47.50	N/A
First-order polynomial convolution interpolation [27]	49.53	N/A

From Fig. 1, it is identified that the recently developed NIL [16] provides higher PSNR than linear interpolation algorithms. Among the linear interpolation algorithms, [11, 13] achieve high PSNR. Figure 2 shows the comparison of PSNR of different cubic-spline interpolation (CSI) methods such as CSI with cross zonal filter (CSICZF) [18, 19], CSI [20], and polynomial spline (PS) [21].

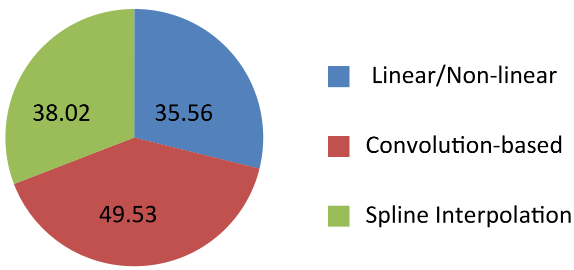
Based on Fig. 2, it is understood that CSIs with CZF [18, 19] provide higher PSNR than simple CSI without filter [20]. However, the PS [21] provides a little bit higher quality as compared with CSIs. Figure 3 shows the comparison of different convolution-based interpolation (CBI) techniques based on the quality of interpolated images. The different CBI techniques are modified convolution-based (MCB) [22], variation framework-based (VFB) [23], second-order polynomial (SOP) [24], modified cubic convolution (MCC) [25], CB with error theorem (CBET) [26], FFOPCI [27], high dimensional model representation with Lagrange (HDMRL) [29], Lagrange with Clamp Filter (LCF) [30] and Bi-cubic (BC) [31].

Based on Fig. 3, the CBET [26] provides higher PSNR by using special error correction scheme. Further, the recently developed interpolation scheme by [29] also provides good interpolated image by producing highest PSNR as 43.77 dB. However, the conventional FOPCI [30] is superior to other CBI schemes. Table 1 summarizes the quantitative measures of different non-adaptive image interpolation algorithms.

From Table 1, it is noticed that the recently developed Lagrange interpolation with HDMR produces a good quality interpolation by providing 44.33 dB PSNR for the interpolated image. However, the convolution-based interpolation algorithms [26, 27] still top in quality. Huang et al. [26] use interpolation error theorem to achieve high-quality convolution-based interpolation. Lin et al. [27] provide an overall higher PSNR as 49.53 dB by using an efficient polynomial convolution interpolation. Based on the comparison of SSIM value of different methods, recently developed non-linear Fuzzy logic-based [16] interpolation method provides higher value than other interpolation techniques. This method provides a high SSIM value as 0.992. Figure 4 shows the overall performance comparison of different kind of non-adaptive interpolation schemes such as non-adaptive linear/non-linear, non-adaptive convolution-based and spline interpolation schemes.

Based on Fig. 4, it is confirmed that the conventional convolution-based interpolation algorithm performs well for producing high-quality interpolated images

Fig. 4 Overall performance comparison of different category of non-adaptive interpolations



by comparing with the other conventional non-adaptive and recently developed non-adaptive image interpolation techniques.

4 Conclusion

This review analyzed the characteristics of different non-adaptive image interpolation algorithms by dividing the algorithms in three major categories such as non-adaptive linear/non-linear, non-adaptive cubic-spline interpolation and non-adaptive convolution-based image interpolation algorithms. Based on the analysis on linear/non-linear interpolation category, this can be concluded that the recently developed non-linear interpolation works better. The review on spline interpolation concludes that the conventional spline interpolation still improves the quality of image as compared with the recently developed cubic-spline interpolation algorithm. The survey on convolution-based image interpolation algorithms suggest that the low-complexity first-order convolution algorithms is very much suitable for producing high-quality images in multimedia devices like digital camera and high definition television and the overall comparison also recommends the first-order convolution for getting better performance in such devices. Further, this survey suggests that all categories of non-adaptive image interpolation algorithms can be improved by using pre-filters to avoid natural image artifacts like blurring and blocking produced by conventional interpolation algorithms and by using simple order approximation for the higher order image scaling algorithms.

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